Yunfei Tian

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Background

Shandong University (SDU) Bachelor 2013.09-2017.06

- B.E. in Optoelec. Info. Sci. and Eng., GPA: 4.35/5.0, Ranking: 3/42.
- Won 2 international honors, 5 national honors, and 6 provincial honors.
- Courses: Digital image processing, geometric optics, microprocessors etc

Nanjing University (NJU) Master 2017.09-2020.06

- M.E. in Opt. Eng., GPA: 4.5/5.0, Ranking: 2/43.
- Research direction: Computational imaging, Super-resolution imaging.
- Published 3 SCI/EI papers and obtained 6 authorized patents.
- Supervisor: Prof. Fei Xu.

Huawei Technologies Co., Ltd. (HUAWEI) Algorithm Engineer 2020.07-2023.07

- Developing, integrating, and implementing image algorithms for classification, detection, segmentation, and tracking in mobile phones or semiconductor factory.
- Real-time object detection in the phone's camera, to help capture clear photos.
- Successfully applied in HUAWEI Phone Mate50, P60 and Mate60 series.
- Work honors include "Tomorrow's Star," "Overcoming Difficulties Award," and "Excellent Team Award" for the Mate50 phone project.

Volkswagen Group China (CARIAD) Algorithm Engineer 2023.09-Present

- Health status contactless detection of drivers and passengers in the car.
- Face detection, facial keypoint detection, and signal processing algorithm development and integration.

Research experience

Super-resolution microscopy imaging NJU 2017.10-2018.12

- Designed and built a low magnification, large field-of-view optical imaging system for microscopy, using multiple light sources to capture multiple low-resolution, large field-of-view intensity images.
- Implemented the Fourier ptychographic imaging algorithm to improve the resolution by more than 10 times. Combined with deep learning-based super-resolution algorithms, further improved the resolution.

Fiber-based single-pixel imaging NJU 2019.01-2020.03

- Used different randomly independent structured light to illuminate the object, and detected the corresponding scattered light power by using a detector. Utilized a fiber-optic head as a single-pixel receiver to achieve remote imaging through fiber-optic transmission.
- Used matrix pseudo-inverse algorithm to calculate the reconstructed object pattern based on the scattered light patterns generated by known multiple LED screens and detected by the detector. Combined with compressed sensing algorithm to reduce the number of detection times and data volume, and speed up the imaging process.

Project experience

Contactless real-time health monitoring in the cockpit
CARIAD 2023.09-present

- Utilizing the cockpit camera to capture facial video signals, through face detection, key point detection, signal filtering, detrending, and Fourier transform to calculate various health indicators such as heart rate, heart rate variability, blood oxygen saturation, respiration rate, etc.
- Supports the analysis report of the driver or passenger's health status, in-car yoga, remote consultation, and other new in-car functions.

Object detection and tracking algorithm of Huawei flagship smartphone camera
HUAWEI 2022.02-2023.07

- As the feature owner, responsible for designing algorithms, model training and quantification, algorithm integration, effect verification, and other related tasks for object detection and tracking in camera preview algorithms.
- Optimize the algorithm to reduce power consumption and load as much as possible, and meet the commercial requirements of smartphone CPU and NPU resources.
- Cooperate with upstream and downstream teams to complete multi-camera mode switching, AWB, AE, AF, APK frame drawing, and other functions.
- Successfully implemented on Huawei's Mate50, P60 and Mate60 phones, supporting Huawei's top-selling features such as long focal length moon mode and 4D focus.

Virtual measurement technology in semiconductor manufacturing
HUAWEI 2021.05-2022.02

- Reduced the frequency of measurements, increased the utilization of measurement equipment, and improved production capacity in a certain semiconductor manufacturing process.
- Abstracted structured data into images and used image interpolation and super-resolution algorithms for reconstruction. Used Attention mechanisms to help users select important measurement points. Also responsible for algorithm service deployment and maintenance.
- Successfully implemented in foundries, breaking the foreign technology monopoly in the field, and applied for an invention patent.

Wafermap image classification in semiconductor manufacturing
HUAWEI 2020.07-2021.05

- Automatic classification of WaferMaps with different patterns can help engineers quickly analyze production status in semiconductor manufacturing.
- Addressed issues such as insufficient data, unlabeled images, annotation errors, multiple labels, and unknown categories using methods such as Autoencoders (AE), Variational Autoencoders (VAE), Transfer Learning, OpenSet Loss, and Center Loss.
- Successfully implemented in foundries, helping engineers locate production faults and establish a smart factory.

Publications

[1]. **Yunfei Tian**, Zixuan Ding, Haogong Feng, Runze Zhu, Ye Chen, Fei Xu*, Yanqing Lu. Single-pixel imaging based on optical fibers[J]. *IEEE Photonics Journal*, 2020, 12(6): 1-7.

[2]. **Yunfei Tian***. Single-pixel imaging by using display illumination[C]. *Real-time Photonic Measurements, Data Management, and Processing IV*. *SPIE*, 2019, 11192: 85-90.

[3]. Danran Li, Nina Wang, Tianyang Zhang, Guangxing Wu, Yifeng Xiong, Qianqian Du, **Yunfei Tian**, Weiwei Zhao, Jiandong Ye, Shulin Gu, Yanqing Lu, Dechen Jiang, Fei Xu*. Label-free fiber nanograting sensor for real-time in situ early monitoring of cellular apoptosis[J]. *Advanced Photonics*, 2022, 4(1): 016001.

Patents

[1]. A numerical aperture measurement method for optical lenses based on computational imaging (201811253669.6, Fei Xu, **Yunfei Tian**, Hui Zou, YanQing Lu, Wei Hu)

[2]. A detection method for identifying surface defects of ultra-thin transparent plates by microscopy (201711145220.3, Fei Xu, **Yunfei Tian**, YanQing Lu, Wei Hu)

[3]. A detection method based on mirror reflection to identify surface defects of ultra-thin transparent plates (201711145236.4, Fei Xu, **Yunfei Tian**)

[4]. A manhole cover anti-theft system based on fiber grating (202022856566.8, **Yunfei Tian**, Lihua Ruan, Shijun Ge, Ye Chen)

[5]. An automatic light tracking device (202020879449.0, **Yunfei Tian**, Lihua Ruan, Shijun Ge, Ye Chen)

[6]. An electromagnetic self-generated fuel-free energy-saving and environmentally friendly lighter (201620104850.0, **Yunfei Tian**, Zhiwei Zhang, Tongfei Zhang)

Skills

- Proficient in programming languages: Python, Matlab, and C++.
- Familiar with deep learning frameworks: PyTorch, TensorFlow, and OpenCV.
- Familiar with tools: Linux, JMESPath, SQL, K8s, Helm, Git, and Django.
- Proficient in software: Zemax, Labview, Photoshop, Premiere, Lightroom, and Office.
- Experienced in SoC development: 51, Msp430, Stm32, Arduino, and Android phones.

Honors and Awards

- National Scholarship (national-level, the ultimate accolade, 2019)
- Second prize in the 10th International College Students ICAN Innovation and Entrepreneurship Competition (international level, 2016)
- Second Prize in China Robot Competition (national-level, 2015)
- First Prize in Shandong Province of Undergraduate Electronic Design Contest (province-level, 2015)
- Outstanding Graduates (university-level, 3/43, 2020)
- The First-class Scholarships (university-level, 2/43, three times, 2017, 2018 and 2019)
- National Inspirational Scholarship (national-level, 2016)